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RAMAIAH Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS – JULY / AUGUST 2025

Program	:	**B.E. – Information Science and Engineering**	**Semester**	:	**IV**
Course Name	:	**Database Management Systems**	**Max. Marks**	:	**100**
Course Code	:	**21IS45 / 22 / 23IS44**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	List any five advantages of using a DBMS over traditional file-based systems.	CO1	(06)
	b)	Discuss the types of languages and interfaces provided by DBMS and the user categories targeted by each interface.	CO1	(06)
	c)	With a diagram discuss the database design process.	CO1	(08)
2.	a)	Explain the Three-Schema Architecture with a diagram.	CO1	(06)
	b)	Illustrate the differences between strong and weak entity sets with examples.	CO1	(06)
	c)	Draw an ER diagram for the MSRM Hospital that requires automation. The business rules are as follows: • A patient can make many appointments with one or more doctors. A doctor can accept appointments with many patients. • Emergency cases do not require any prior appointments. However these cases must be recorded in the database as "unscheduled". • Every visit yields a diagnosis followed by an appropriate treatment. • Each visits updates the patient's records to provide medical history • Each patient visit creates a bill and this bill is linked to particular doctor. Also each doctor can bill many patients. Bills may be paid in cash, or through a credit-card, or claimed through "mediclaim".	CO1	(08)

UNIT - II

3.	a)	Explain the difference between external, internal, and conceptual schemas. How are these different schema layers related to logical and physical data independence?	CO2	(06)
	b)	Consider the following relational database schema consisting of the four relation schemas: **passenger** (pid, pname, pgender, pcity) **agency** (aid, aname, acity) **flight** (fid, fdate, time, src, dest) **booking** (pid, aid, fid, fdate) Answer the following questions using relational algebra queries: i. Get the complete details of all flights to New Delhi. ii. Get the details about all flights from Chennai to New Delhi.	CO2	(10)

iii. Find only the flight numbers for the passenger with pid 123 for flights to Chennai before 06/11/2020.
iv. Find the passenger names for those who do not have any bookings on any flights.
v. Get the details of flights scheduled on both dates, 01/12/2020 and 02/12/2020, at 16:00 hours.

c) List and explain the characteristics of the Relational Model. CO2 (04)

4. a) Perform the following operations on the two relations R and S as given: Cartesian Product, Join, Union, Intersection. CO2 (08)

Employee

ID	NAME
101	Sachin
103	Rahul
104	Kapil
107	Ajay

Salary

ID	SALARY
101	65000
103	35000
104	22000
107	21910

b) Consider the following relations R and S: CO2 (08)

R

B	C	D
5	2	6
3	3	5
4	5	1

S

A	B
2	3
6	4
3	5
3	6

What will be the result after the following Relational Algebra Operations?

i) R ⋈ S
ii) R ⋈ R.B < S.D AND R.A
iii) π B, C (R ⋈ S)
iv) R ⋈ R.B > S.D AND R.A

c) What is the difference between a candidate key and the primary key for a given relation? Illustrate with an example. CO2 (04)

UNIT - III

5. a) Explain the different joins supported in SQL with examples CO3 (05)

b) The following tables are maintained by a book dealer. CO3 (08)

AUTHOR (author-id:int, name: string, city: string, country: string)
PUBLISHER (publisher-id:int, name: string, city: string, country: string)
CATEGORY (category-id:int, description: string)
CATALOG (book-id:int, title: string, author-id:int, publisher-id:int, category-id:int, year:int, price:int)
ORDER-DETAILS (order-no:int, book-id:int, quantity:int)

Write the following queries in SQL:

(i) Give the details of the authors who have 2 or more books in the catalog and the price of the books is greater than the average price of the books in the catalog and the year of publication is after 2000.
(ii) Find the author of the book which has maximum sales.
(iii) Demonstrate how you increase the price of books published by a specific publisher by 10%.

c) Explain how ***Group By*** clause works. What is the difference between ***Where*** clause and ***Having*** clause? CO3 (07)

6.	a)	With an example, explain the need and syntax of SET operation in SQL.	CO3	(06)
	b)	For below database keeps track of auto sales in car dealership, Solve these queries using SQL. emp(Name, ssn, Address, sex, salary, dno) dept(Dname, Dnum, Mgrssn, Mgrstartdate) dept_loc(Dnum, Dloc) project(Pname, Pnum, Ploc, Dnum) works_on(essn, pno,hrs) dependent(essn, Dep_name, Sex, Bdate, Relation) (i)Retrieve all employees whose address is in 'Bangalore'. BRARY (ii)Show the resulting salaries if every employee working on the 'aaa' project is given a 10 percent raise. iii)Display dnum, name of an emp, and pno for all the employees who work on projects which belongs to their department. iv)Display all the employees who have 2 or more dependents	CO3	(08)
	c)	Explain different Schema change statements in SQL.	CO3	(06)

UNIT- IV

7.	a)	What is the purpose of normalization? Explain the First and Second Normal forms with examples.	CO4	(06)
	b)	Write the algorithm to find the closure of functional dependencies. Let F={ssn⟶Ename}, Pnumber⟶{Pname,Plocation},{ssn,Pnumber}⟶Hours}. Calculate the following closure sets with respect to F. $\{Pnumber\}^+$,$\{ssn,Pnumber\}^+$	CO4	(07)
	c)	Why should NULLs in a relation be avoided as far as possible? Discuss the problem of spurious tuples and how we may prevent it.	CO4	(07)
8.	a)	Explain with an example for the Informal guidelines for a relation schemas.	CO4	(07)
	b)	List the Inference rules for functional dependencies. Write the algorithm to determine the closure of X (set of attributes) under F (set of functional dependencies).	CO4	(06)
	c)	Explain BCNF. State if the following relation is in BCNF. If not, decompose the same into BCNF. State the candidate key.	CO4	(07)

STUDENT	COURSE	INSTRUCTOR

FD1: {Student, Course} -> Instructor
FD2 -> Instructor -> Course

UNIT - V

9.	a)	Differentiate between single-user and multi-user transactions.	CO5	(06)
	b)	Describe the recovery technique based on Deferred Update.	CO5	(06)
	c)	Explain Transaction and System Concepts. Discuss the role of Read and Write operations within a transaction.	CO5	(08)
10.	a)	List and explain any four desirable properties of transactions (ACID).	CO5	(06)
	b)	What is MongoDB? Explain its key features and advantages over traditional relational databases.	CO5	(06)
	c)	Explain the characteristics and types of NoSQL databases. How is NoSQL used in the industry today?	CO5	(08)

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